

Medical Intent Annotation Guideline (per-Sentence)

The primary goal of this guideline is to help annotators accurately label the **intent behind each sentence** in medical **release notes** or **clinical visit summaries**. These notes are written by healthcare professionals **after a patient's appointment**, and they summarize the entire visit based on the doctor's interaction with the patient.

Each note includes a variety of medical content such as:

1. The patient's **chief complaint**
2. The doctor's **observations and findings**
3. **Diagnoses** made during the visit
4. **Treatments** or **medications** prescribed
5. **Tests** or **referrals** ordered
6. Any **discussions or advice** shared with the patient

The sentences you will annotate reflect the doctor's clinical finding, reasoning, actions, and recommendations. They are not direct dialogue, but rather a professional summary of what occurred during the appointment and what decisions were made.

Your task is to assign **one intent per sentence** that best describes the **purpose** or **function** of that sentence. This could include intents like *acute symptoms*, personal history, *diagnostic testing*, *medication*, *discussion*, or *reassessment*, among others.

Please make sure to:

1. Read each sentence **carefully in context**
2. Choose the **most appropriate intent**
3. **Stick strictly to the annotation guidelines**
4. Avoid assumptions beyond what is written

Accurate intent annotation is essential for creating high-quality data that will be used for further research. Your attention to detail and consistency is key to the success of this project.

Data Information

The data is obtained from: [Aci-bench: a Novel Ambient Clinical Intelligence Dataset for Benchmarking Automatic Visit Note Generation | Scientific Data](#)

The whole note looks like this in the dataset. However, **your task is not to annotate the entire note at once**, but rather to **assign an intent to each sentence individually**. The structure of these notes generally follows a **consistent pattern across all cases**. This consistent format should help you better understand the role each sentence plays, making it easier to determine the correct intent.

-----Example Note-----

-CHIEF COMPLAINT

Back pain.

-HISTORY OF PRESENT ILLNESS

Mr. John Perry is a 61-year-old male with a past medical history significant for kidney stones, migraines, and gastroesophageal reflux, who presents with some back pain. The patient reports that he is feeling a lot of the same pain that he had when he had kidney stones about 2 years ago, so he is a little concerned. The pain started from the right side and moved over and he feels it on the left side of his back. This has been going on for the last 4 days. Initially, the pain was intermittent, but over the last 48 hours it has been constant. He has taken Tylenol, but it does not seem to help. He thinks he has hematuria, but it is hard to detect but it does look a little off color. He endorses nausea and vomiting if he exerts himself or climbs the stairs to his apartment or runs to catch the bus. He also endorses dizziness and lightheadedness with pain in his abdomen.

Regarding his migraines, he has been diligent about taking the Imitrex. He denies issues with the migraines.

Regarding his gastroesophageal reflux, he reports that he has been doing well with his diet, but notes lately with his pain, he has been eating more fast food and takeout since these options come with delivery. He is staying hydrated. He is taking Protonix 40 mg daily as directed.

-REVIEW OF SYSTEMS

â€¢ Gastrointestinal: Endorses abdominal pain. Endorses nausea and vomiting with exertion.

â€¢ Genitourinary: Endorses urine discoloration.

â€¢ Musculoskeletal: Endorses back pain. Endorses body aches.

â€¢ Neurological: Denies headaches. Endorses dizziness and lightheadedness.

-PHYSICAL EXAMINATION

â€¢ Respiratory: Lungs are clear to auscultation bilaterally. No wheezes, rales, or rhonchi.

â€¢ Cardiovascular: No murmurs, gallops, or rubs. No extra heart sounds.

â€¢ Gastrointestinal: Tender to palpation to the right lower quadrant. CVA tenderness on the right.

-VITALS REVIEWED

â€¢ Blood Pressure: Elevated.

-RESULTS

Creatinine level slightly elevated.

Abdominal x-ray demonstrates possible kidney stone.

-ASSESSMENT AND PLAN

Mr. John Perry is a 61-year-old male with a past medical history significant for kidney stones, migraines, and gastroesophageal reflux, who presents with back pain.

Kidney stones.

â€¢ Medical Reasoning: He is experiencing pain in his back that is similar to his previous kidney stone pain. His recent abdominal x-ray demonstrates what appears to be a recurrent kidney stone.

â€¢ Additional Testing: I have ordered a CT scan of the abdomen and pelvis without contrast.

â€¢ Medical Treatment: We will start him on Ultram 50 mg as needed every 6 hours for pain.

â€¢ Patient Education and Counseling: I advised the patient to stay well hydrated and to strain his urine.

Migraines.

â€¢ Medical Reasoning: He has been compliant with Imitrex and is doing well at this time.

â€¢ Medical Treatment: Continue Imitrex.

Reflux.

â€¢ Medical Reasoning: This is typically well-controlled with dietary modifications.

â€¢ Medical Treatment: Continue with Protonix 40 mg daily. A refill was provided.

-Patient Agreements: The patient understands and agrees with the recommended medical treatment plan.

This is how the sample will look while annotating. Instead of the entire note, every window will have one of the SOAP blocks for easier understanding.

-----example sample----- CHIEF COMPLAINT

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-HISTORY OF PRESENT ILLNESS

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Annotation Class

The orientation of the note follows the SOAP principle which represents **Subjective**, **Objective**, **Assessment**, and **Plan**. The note will always follow this structure and the information will travel through subjective to plan sequentially. In this annotation guideline these 4 class will have in total 18 sub classes following some good examples of the samples for each class.

Subjective Intent

This is the first heading of the SOAP note. Documentation under this heading comes from the “subjective” experiences. In the note, this part start with *Chief Complaint (CC)*. Which is the problem reported by the patient at first. This can be a symptom, condition, previous diagnosis or another short statement that describes why the patient is presenting today and this kind of represent the whole scenario.

Personal information

Normally, it refers to the greeting but in note there is no direct greetings term. Mostly it starts with mentioning the name and age of the patient and finishes up the reason of the visit.

- example sample-----
- a. Ms. Thompson is a 43-year-old female who presents today for an evaluation of right knee pain
 - b. "Raymond Taylor is a pleasant 67-year-old male who presents to the clinic today for difficulty swallowing."
 - c. Natalie Price is a 39-year-old female here today for a checkup for an abnormal labs.
 - d. Maria Martin is a 75-year-old female who presents today for follow-up for diabetes.
-

Acute symptoms

This section is the detailed account of the patient's chief complaint, based on the discussion during the appointment. In this section the doctor highlight the current symptoms and it might start with When did the CC begin following Where, how long, Characterization, and intensity. Key elements typically included in this part of the note are:

- **Onset of Symptoms:** The exact or approximate time when the symptoms first appeared.
- **Location of Symptoms:** The specific area of the body affected.
- **Duration:** How long the symptoms have been occurring.
- **Characterization of Symptoms:** A description of the symptoms (e.g., sharp pain, throbbing sensation, intermittent discomfort).
- **Intensity of Symptoms:** Often assessed on a scale from mild to severe, indicating the severity of the condition at the time of the visit.

- example sample-----
- e. "The onset of her pain began 7 days ago, when she was playing basketball and another player ran into her."
 - f. The patient reports that about a week ago he started to get a cold, which has worsened in the last 2 or 3 days.
 - g. She reports these symptoms while walking a mile, going to work, and performing daily tasks.
 - h. The patient was doing yard work over the weekend when he began to feel short of breath and a full chest pain for approximately hour.
 - i. The pain is constant. He states the pain is primarily located in the left shoulder and denies it radiates down into the left arm.
-

Personal History

The "personal history" intent encompasses all inquiries regarding a patient's past medical events. This includes questions about chronic conditions such as diabetes and checks on vital signs like heart rate. It also covers inquiries about symptoms experienced in the past that may be relevant to the patient's current complaint. Additionally, this intent includes questions about the patient's diet and allergies, as it can provide essential insights into their overall health and specific medical conditions.

- example sample-----
- a. "The patient reports she is a diabetic who takes her blood sugar every morning."
 - b. "Patient reports history of diabetes and heart disease."
 - c. Regarding his congestive heart failure, he has been watching his diet and trying to exercise regularly.
 - d. The patient underwent right knee surgery a couple of months ago and has been feeling well since then.
 - e. Patient reports history of history of hypertension, diabetes, and kidney stones.
 - f. ALLERGIES Penicillin.
-

Family History

This intent focuses on evaluating the patient's family medical history. It typically involves direct inquiries about similar symptoms or health conditions experienced by relatives.

-----example sample-----

- a. "Patient's father reports that they do not have a family history of seizures or Tourette's syndrome."
- b. "Patient reports both of her parents have hypertension and one also had kidney disease."
- c. There is a family history of melanoma in her mother and maternal aunt.(D2N028)
- d. He denies a family history of heart disease.
- e. He reports a family history of high blood pressure, diabetes, and depression

Drug History

The "drug history" intent records both current and past use of non-prescribed substances by the patient. This includes alcohol, caffeine, nicotine, cannabis, and other illicit drugs.

-----example sample-----

- a. The patient states that he drinks 2 beers once a week and denies a history of heavy alcohol or drug use (D2N028)
- b. She continues to smoke but has decreased from two packs a day down to a couple of cigarettes daily.
- c. Former smoker. Quit approximately 30 years ago.
- d. He notes he used to smoke more and is having difficulty with complete cessation.
- e. Alcohol: Socially. No history of heavier consumption.

Therapeutic History

The "therapeutic history" or "medication history" intent involves a doctor inquiring about any medications or therapies the patient has used in the past or is currently using. It can also involve any other form of treatment the patient applies in the past and present rather than medication or therapies. This information helps the doctor understand previous and ongoing treatments to manage the patient's condition effectively.

-----example sample-----

- a. "Currently on Lisinopril for hypertension and Topamax for migraine prevention."
- b. Regarding her depression, she reports that she has been going to therapy every week for the past year. (D2N001)
- c. The patient has taken Tylenol and iced his knees for a short period of time, but nothing really seemed to help. (D2N002)
- d. Regarding his migraines, he has been diligent about taking the Imitrex
- e. He has been taking ibuprofen, which has not been beneficial alone.

Vegetative History

The "vegetative history" intent refers to documenting inquiries and observations about a patient's basic internal body functions. This includes recording details such as fatigue and other key points typically addressed in a review of systems, providing a comprehensive overview of the patient's physiological state.

-----example sample-----

- a. "Raymond Taylor is a pleasant 67-year-old male who presents to the clinic today for difficulty swallowing."
- b. He states he was nauseous for a couple of days but that was because he was sitting in the back of a car
- c. The patient denies chest pain, shortness of breath, vomiting, dizziness, fevers, and chills.

REVIEW OF SYSTEMS

â€¢ Constitutional: Denies fevers, chills, or weight loss. Endorses fatigue.

â€¢ Ears, Nose, Mouth and Throat: Endorses nasal congestion from allergies.

â€¢ Cardiovascular: Denies dyspnea.

â€¢ Respiratory: Denies cough, shortness of breath.

â€¢ Gastrointestinal: Denies hematochezia, melena.

â€¢ Musculoskeletal: Endorses bilateral leg swelling.

â€¢ Neurological: Endorses dizziness.

Other Social:

The "other socials" intent serves as a broad category for documenting information about a patient's current social circumstances and external factors that might impact their health. This includes details about their family status, such as children and marital status, employment, living conditions, available social support networks, involvement in sports, and other relevant social activities.

- example sample-----
- a. "Patient reports that she plays basketball at the Y for fun."
 - b. This has been more of a struggle recently as she has been traveling a lot.
 - c. The patient exercises in the morning and rides a bike for 45 minutes.
 - d. She is no longer doing yoga as she has not made it a habit.
 - e. Patient owns an auto repair shop.
-

Objective Intent

This section documents the objective data from the patient encounter. This provides a concise, yet detailed record of factual, measurable data observed during a patient's appointment. It includes vital signs such as blood pressure, temperature, and heart rate, detailed findings from the physical examination (e.g., heart sounds, lung function, examination of the abdomen), and concrete data from diagnostic tests like blood tests or imaging results.

Physical Examination

The "physical examination" intent refers to the documentation of the doctor performing hands-on physical tests on the patient during the appointment. This includes observing, touching, or using instruments to assess different parts of the body. It is typically clear and direct, and includes general appearance, specific system evaluations (like respiratory or cardiovascular), and any signs of pain, abnormal findings, or limitations in movement. The language is factual and precise, capturing what the doctor directly observed or measured during the exam.

- example sample-----
- a. "MSK: Examination of the right knee: No pain to palpation of the lateral aspect of the right knee."
 - b. "CV: 2/6 systolic ejection murmur heard at the left base."
 - c. Musculoskeletal: There is edema and erythema of the right knee with pain to palpation.
- Vital Signs: (Physical examination)
"Blood pressure: 127/80"
"Heart rate: 60s"
-

Lab Examination

This intent refers to the part of the medical note where the doctor reviews and discusses test results that were performed in a laboratory. These may include blood tests, urine tests, or other medical analyses that help evaluate the patient's health. In the note, this intent is shown when the doctor refers to these test results as evidence to explain symptoms, confirm a diagnosis, check for abnormalities, or monitor how well a treatment works.

- example sample-----
- a. White blood cell count is within normal limits
 - b. Shave biopsy of right inferior back malignant melanoma, invasive, superficial spreading.
 - c. Rapid strep test: positive
 - d. Glucose and hemoglobin A1c elevated
 - e. Non-fasting glucose is elevated.
 - f. Salivary Cortisol Tests: 2 out of 3 were normal. Not consistent with Cushing's Syndrome.
-

Radiology Examination

It refers to the part of the medical note where the doctor reviews and interprets imaging studies done for the patient. These screening procedures may include X-rays, MRI scans, CT scans, ultrasounds,

echocardiograms, and other imaging tests. This section provides information about what was found in the images and how those findings relate to the patient's symptoms or condition.

- example sample-----
- a. X-ray of the right ankle taken in office today reveals no fracture or bony abnormalities.
 - b. X-rays of the right knee taken in office today reveal the bony alignment in good position.
 - c. These reveal subtle dorsal displacement of the base of the 2nd metatarsal with a 3 mm separation of the 1st and 2nd metatarsal bases.
 - d. Electrocardiogram is reviewed and revealed normal sinus rhythm with good R wave progression and evolutionary changes, which are anticipated.
 - e. CT scan of the abdomen was reviewed today and revealed a kidney stone measuring 0.5 cm in size, located in the proximal right ureter.
-

Assessment Intents

The **Assessment** section is where the doctor explains their medical opinion based on what the patient said (subjective data) and what was observed or measured (objective data). It includes a summary of what happened during the visit and compares the patient's condition to previous visits—whether they are getting better, worse, or staying the same. The doctor may also point out any patterns or things that don't match up. If there are any challenges that could affect the patient's progress, like stress, memory problems, or home environment, they are also mentioned here. This section helps make sense of all the information and leads to a diagnosis or next steps.

Acute Assessment

The "acute assessment" intent is used when the doctor analyzes the patient's current symptoms and draws a clear, primary diagnosis based on the information gathered during the visit. This includes the patient's reported complaints, physical examination findings, and any relevant test results. The purpose of this intent is to identify the immediate cause of the patient's current health issue and determine the appropriate next steps for treatment or management. It often appears in cases where the patient visits with a new or sudden-onset problem, such as pain, fever, or injury. In Note, it can start with Medical reasoning.

- example sample-----
- a. Given that his right knee x-ray was unremarkable, I believe this is an acute exacerbation of his arthritis.(D2N002)
 - b. Right middle finger distal phalanx fracture.
 - c. I believe his symptoms which led to his ED visit were due to an exacerbation of his heart failure
 - d. His hemoglobin was 8.2, which is low for his height and weigh
 - e. Endoscopy from the hospital showed gastritis which could be the source of bleeding
 - f. The biopsy revealed an intermediate thickness melanoma
-

Reassessment

This intend is used when the doctor evaluates a diagnosis that is already known to the patient. It applies to chronic conditions and follow-up visits, where the focus is on monitoring the stability, progression, or control of an existing illness. Instead of identifying a new problem, the doctor reviews how a previously diagnosed condition—such as diabetes, hypertension, or asthma—is currently being managed and whether any changes to the treatment plan are needed.

- example sample-----
- a. Medical Reasoning: He is doing well from this standpoint. He has been compliant with his medications, dietary modifications, and regular physical activity.
 - b. Her echocardiogram from 6 months ago demonstrated a reduced ejection fraction of 45%, as well as moderate mitral regurgitation. .(D2N002)
 - c. He is doing well with therapy and physical activity
 - d. The patient is doing well on Synthroid and is asymptomatic at this time.
 - e. Physical therapy has also been beneficial for her
 - f. He saw his neurologist about 3 months ago and has been asymptomatic for the past few months.
-

Plan Intents

The plan part of the note is based on the findings from the assessment and outlines what should be done to manage, treat, or further investigate the condition.

A **plan** can include multiple actions, such as ordering lab tests or imaging, prescribing medications, monitoring the condition over time, or referring the patient to a specialist. It may also involve giving the patient education to help them understand their condition and how to care for themselves.

Medications

This part involves direct mention of medications for the patient to take. This includes details such as the name of the medication, dosage, frequency, and duration. The purpose of this intent is to treat or manage the patient's condition through drug therapy as part of the overall care plan.

-----example sample-----

- a. "I have prescribed the patient meloxicam 15 mg once a day to treat the pain and swelling."
- b. "I have prescribed the patient Protonix 40 mg first thing in the morning to treat his acute esophagitis."
- c. Medical Treatment: Continue lisinopril 20 mg once daily and Lasix. Lisinopril was refilled today.
- d. Continue omeprazole 40 mg once daily.

Diagnostic Testing

The **"diagnostic testing" intent** is used when the doctor orders additional medical tests to gather more information about the patient's condition. This can include laboratory tests (such as blood or urine analysis), imaging procedures (like MRI, ECG, or X-ray), and basic measurements (such as blood pressure or glucose levels). It's like repeating the objective part based on the current assessment. These tests help clarify the diagnosis, monitor the patient's status, or guide the next best steps in treatment.

-----example sample-----

- a. Additional Testing: We will order a lipid panel and an echocardiogram.
- b. Additional Testing: We will order a repeat echocardiogram. We will also repeat a lipid panel this year.
- c. Additional Testing: We will order an autoimmune panel for further evaluation
- d. Additional Testing: I have ordered a CT scan of the abdomen and pelvis without contrast.
- e.

Other Treatments

Some treatments do not fit into the standard categories like medication or diagnostic testing. In such cases, the **"other treatment" intent** is used to document medical actions such as recommending orthopedic aids (e.g., crutches or slings), advising on diet or lifestyle improvements, or planning for surgery. This intent ensures that all types of therapeutic measures are captured, even if they are not specifically detailed elsewhere in the documentation.

-----example sample-----

- a. I have recommended we place her into a long arm cast and sling. At that point, we may need to discuss surgical options. (D2N015)
- b. I have recommended a sling for comfort
- c. I encouraged the patient to reduce his caffeine consumption avoid NSAIDs and alcohol.
- d. The patient will also continue working on weight loss, which should additionally improve her liver enzymes
- e. I also recommend that the patient report to occupational therapy once a week for the next 6 weeks to work on full active and passive right index finger range of motion with no restrictions.

Discussion

This intent is used when the doctor includes a note about discussing the treatment plan with the patient to address any concerns or questions. It indicates that, after outlining the next steps, the doctor took time to explain the reasoning, clarify instructions, and ensure the patient understood the plan. This part of the note reflects shared decision-making and supports patient understanding and involvement in their care.

- example sample-----
- a.

"Regarding her diabetes, she should continue to monitor her blood sugars daily."
- b.

"We discussed that his pumping function should improve in time."
- c.

The patient understands and agrees with the recommended medical treatment plan.
- d.

I advised the patient to discontinue the use of ibuprofen, but he may continue using Tylenol if he wishes.
-

Follow-up

After outlining the treatment plan, the doctor may schedule a follow-up appointment to monitor the patient's progress. This includes checking whether symptoms have improved or changed, and evaluating the effects of any medications or therapies given and it will be directly mention in the note.

- example sample-----
- a.

She will follow up in 2 weeks for repeat imaging. (D2N015)
- b.

If his symptoms persist, we can consider further imaging and possibly a referral to physical therapy.
- c.

She will follow up in 2 weeks for repeat imaging.
- d.

She will follow up with me as needed if she continues to have pain
- e.

Follow up in 6 months to reevaluate scars.
-

Referrals

When the doctor decides that further evaluation or treatment is needed beyond their expertise, a referral is made to another specialist or healthcare provider. It will be directly mentioned at the end of the note.

- example sample-----
- a.

Specialist Referrals: We will refer him to physical therapy to work on strengthening the area.
- b.

Specialist Referrals: We will refer him to physical therapy to be started after we get his MRI results back.
- c.

Currently he is not seen by cardiology routinely and I have suggested he follow up with Dr. Tyler Sanchez.
- d.

Schedule a follow up appointment in 3 months to assess her pain.
- e.

The patient will follow up in 2 weeks.
-

Similar-Sounding Intents and How to Distinguish Them

In clinical notes, some intents may look or sound similar, especially when a sentence refers to symptoms, past conditions, or assessments. Below are common pairs that may cause confusion, along with guidelines to separate them accurately.

Acute Symptoms vs. Personal History

Why they seem similar: Both can describe symptoms the patient has experienced.

Intent	Description	Clue Words
Acute Symptoms	Recent or currently active symptoms	"complains of", "has had for the past few days", "recently developed"

Personal History	Past medical events or long-standing conditions	<i>"has a history of...", "10 years ago", "previous surgery"</i>
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Acute Assessment vs. Reassessment

Why they seem similar: Both involve the doctor making an assessment or clinical judgment.

Intent	Description	Clue Words
Acute Assessment	A new problem is being assessed based on current symptoms	<i>"Presents today with...", timeline like "for 3 days", "new onset"</i>
Reassessment	A known or chronic condition is being re-evaluated	Mentions of stable/chronic conditions like <i>"diabetes, hypertension"</i> , or phrases like <i>"has been doing well with..."</i>

Special Case: During routine or annual visits, patients typically do not present with a new complaint. In these cases, there is usually no acute assessment. Instead, the focus shifts to a reassessment of known, ongoing conditions such as diabetes, hypertension, or asthma. The doctor reviews the stability or progression of these chronic issues, ensuring they remain under control and updating the care plan if needed.

Discussion vs. Other Plan Intents (e.g., Medication, Referral, Diagnostic Testing)

These intents can easily overlap because they often appear in the same section of the note. However, the purpose of the sentence is what sets them apart.

The Discussion intent captures any part of the note where the doctor is:

- ✓ **Explaining** the diagnosis or plan to the patient
- ✓ **Giving options** or conditional suggestions
- ✓ **Answering or anticipating questions**
- ✓ **Ensuring patient understanding or agreement**
- ✓ **Clarifying** instructions, risks, or next steps

Key Clues:

- ❖ Words like *"discussed", "talked about", "agreed to", "if symptoms persist...", "the patient understands...", "we reviewed..."*
- ❖ Sentences that reflect **conversation** or **shared decision-making**

Plan intents (like **Medication, Referral, Diagnostic Testing, Follow-up**) are used when the doctor is:

- ✓ Making a **clear medical decision**
- ✓ Giving a **direct order** or instruction
- ✓ Outlining an **action to be taken**

Key Clues:

- ❖ Sentences that sound like a **prescription, referral, test order, or next appointment**
- ❖ No language suggesting a **conversation** or **patient involvement**